

Highlights

Agri-MetaRL: An Agricultural Meta-Reinforcement Learning Algorithm for Greenhouse Climate Control

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- Greenhouse control is modeled as a task distribution over weather and scenario.
- MetaAdvantageHead on Recurrent PPO for task-adaptive advantage correction.
- Evaluation protocol with training and held-out testing is constructed.
- Agri-MetaRL achieves highest EPI (3.46 €/m²) under the current protocol.
- The method shows competitive temperature control and held-out performance.

¹These authors contributed equally to this work.

Agri-MetaRL: An Agricultural Meta-Reinforcement Learning Algorithm for Greenhouse Climate Control

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Abstract

Greenhouse climate control aims to ensure suitable growing conditions and reduce energy consumption. Greenhouse systems exhibit strong nonlinearity, time-varying disturbances, and changing crop stages, which make traditional methods (PID, rule-based control, model predictive control) prone to failure under cross-season and cross-weather conditions. Reinforcement learning (RL) offers a data-driven path, but most studies train a single policy and pay insufficient attention to cross-task transfer. This paper formulates greenhouse control as a set of related tasks defined by weather year, start date, and scenario conditions. It proposes Agri-MetaRL, a meta-reinforcement learning algorithm that improves advantage estimation at the algorithmic level. Based on Recurrent PPO, Agri-MetaRL introduces a support-query advantage correction mechanism and a custom module, MetaAdvantageHead, which encodes task context from support trajectories and performs task-adaptive cor-

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rection on query trajectories. The design avoids extra policy branches and heavy bi-level optimizers. Experiments on the GreenLight-Gym benchmark under 5 random seeds show that Agri-MetaRL achieved a final mean training return of 3955, higher than Recurrent PPO (3886) and PPO (3811). Under the fixed protocol, Agri-MetaRL had the lowest violation steps for temperature, humidity, and CO₂, and the highest EPI of 3.46 €/m². Code is available at <https://github.com/1240945123/Agri-MetaRL>.

Keywords: Greenhouse climate control; Reinforcement learning; Meta-reinforcement learning; Recurrent PPO; Smart agriculture

1. Introduction

Protected agriculture provides stable growing conditions for crops by regulating temperature, humidity, CO₂ concentration, and radiation. This approach can increase yield per unit area, extend supply periods, and resist extreme weather^{? ?}. In modern high-tech greenhouses, environmental variables, crop state, and actuators are strongly coupled. External weather and crop growth stages change continuously. Thus, greenhouse control has characteristics of strong time-variation, partial observability, and multi-objective optimization. How to achieve high returns, low energy consumption, and strong generalization control while ensuring crop safety bounds is one of the core problems at the intersection of agricultural engineering and intelligent control.

Traditional greenhouse control often uses PID, threshold rules, and manual experience-based strategies^{? ?}. These methods are simple to implement and highly interpretable. But parameters are usually tuned for specific sea-

16 sons, specific greenhouses, and specific crop stages. When weather distri-
 17 bution changes or crop state shifts, re-tuning is often required. Model pre-
 18 dictive control (MPC) can use future predictions and explicit constraints to
 19 improve control quality[?]. But MPC performance depends on high-precision
 20 greenhouse dynamics models and stable solving processes. When model er-
 21 ror, parameter uncertainty, and long-horizon rolling optimization coexist, de-
 22 ployment cost remains high. This has been reflected in studies on economic
 23 greenhouse MPC and full-cycle crop setpoint optimization^{???}. RL-guided
 24 MPC and RL-based MPC have been proposed as compromise solutions to
 25 balance learning ability and optimization structure^{??}.

26 In recent years, reinforcement learning (RL) has received wide attention.
 27 RL does not rely on precise analytical models and can directly optimize
 28 policies toward cumulative returns^{??????}. GreenLight-Gym provides
 29 a standardized high-fidelity benchmark environment for greenhouse RL[?].
 30 Greenhouse RL and deep RL control have been used in recent years for
 31 comparing RL with MPC, improving energy efficiency, and introducing in-
 32 teractive control^{??}. But most existing work limits training to a single or
 33 nearly single environment distribution. As a result, the learned policies are
 34 often closer to “a strong controller trained on one representative task” than
 35 to “a task-aware update structure that transfers across a family of related
 36 tasks.”

37 From a learning perspective, changes in weather year, start planting date,
 38 parameter perturbations, and scenario coefficients make greenhouse control
 39 better viewed as a family of related tasks rather than a single MDP. The goal
 40 of meta-reinforcement learning (Meta-RL) is not only to learn a static opti-

mal policy, but also to learn an update structure that transfers across related tasks. Meta-gradient RL, distributional meta-gradient RL, first-order meta-learning, and comparisons of fine-tuning with Meta-RL all show that learning update rules or adaptation priors can significantly affect cross-task performance. Inspired by this line of work, this paper proposes a new meta-reinforcement learning algorithm, Agri-MetaRL, and formulates greenhouse control as a Meta-RL problem over weather year, start date, and scenario conditions. The contribution of this paper lies in algorithm design, not merely in applying off-the-shelf RL to greenhouse scenarios. Instead of adding multi-policy branches or constructing heavy inner-outer optimizers, this paper performs task-adaptive correction on advantage estimation. Specifically, task context is extracted from support trajectories, and corrected advantages are generated on query trajectories to guide PPO policy updates. This design aligns with the research thread of improving RL algorithms by learning optimization objectives or update rules.

Unlike the traditional “single-environment RL + test-period extrapolation,” this paper introduces a task-partitioned training skeleton at the implementation level. Each task segment in the training buffer is split into support and query sets. The support set encodes context. The query set computes task-adaptive advantages for the auxiliary correction module. This design brings Agri-MetaRL closer to task-conditioned policy optimization while retaining training-flow compatibility with standard PPO and engineering simplicity.

The main contributions of this paper are as follows:

1. For greenhouse climate control, a task-distribution modeling approach

over weather year, start date, and scenario conditions is proposed, together with corresponding training and testing protocols for agricultural meta-reinforcement learning.

2. A new meta-RL algorithm design is proposed: task-adaptive correction is introduced at the advantage estimation step, rather than merely applying off-the-shelf RL to greenhouse scenarios.
3. The custom module MetaAdvantageHead is designed and implemented. It performs lightweight task-conditioned advantage correction through support-set context encoding. This module differs from standard Recurrent PPO and is a researcher-developed algorithm component.
4. On the GreenLight-Gym benchmark, a unified evaluation matrix is constructed including learning curves, climate bounds, economic metrics, and held-out evaluation.

2. Materials and Methods

2.1. Simulation Environment

Experiments in this paper were conducted on the GreenLight-Gym greenhouse reinforcement learning platform^{???}. The platform exposes high-fidelity greenhouse climate dynamics, crop growth, and actuator control through the Gymnasium interface. It supports flexible configuration of weather, parameters, and observation modules. The simulation object is a Venlo-type glass greenhouse common in the Netherlands, with tomato (*Solanum lycopersicum*) as the crop. External weather uses Amsterdam historical weather data. Each episode represents a 60 d growth window. The control step is 15 min. Thus each episode contains 5760 control steps.

90 *2.2. Observation and Action Spaces*

91 The agent’s observation vector has 263 dimensions. It consists of six types
92 of information, as shown in Table 1.

Table 1: Composition of the agent observation vector in the GreenLight-Gym tomato greenhouse environment

Component module	Dimension
Indoor climate	4
Crop state	3
Control signals	6
Current weather	5
Time encoding	5
Weather forecast	240

93 Observations include indoor climate, crop state, current control inputs,
94 current weather, time encoding, and weather forecasts for the next 48 time
95 steps. Running mean and variance were used to normalize observations dur-
96 ing training. The action space is a 6-dimensional continuous vector. It
97 corresponds to six actuator types: heating, CO₂ injection, thermal screen,
98 ventilation, supplemental lighting, and shade screen. The action range is
99 $[-1, 1]$. The system updates control inputs in an incremental manner:

$$u_{t+1} = \text{clip}(u_t + a_t \cdot \Delta u_{\max}, u_{\min}, u_{\max}), \quad (1)$$

100 where $\Delta u_{\max} = 0.1$. The incremental control mechanism can reduce abrupt
101 actuator changes and improve control smoothness and training stability.

102 2.3. Reward Function and Constraints

103 This paper adopts the profit-oriented reward function in GreenLight-
104 Gym. The single-step reward is defined as

$$r_t = \Delta m_{\text{fruit}} p_{\text{fruit}} - (c_{\text{heat}} + c_{\text{CO}_2} + c_{\text{elec}}) - \boldsymbol{\lambda}^\top \mathbf{v}_t, \quad (2)$$

105 where Δm_{fruit} denotes fruit mass increment, $p_{\text{fruit}} = 1.6 \text{ € kg}^{-1}$ is the tomato
106 price, c_{heat} , c_{CO_2} , and c_{elec} denote heating, CO₂, and electricity costs, respec-
107 tively, and $\mathbf{v}_t$ denotes the constraint violation amount. The safe ranges for
108 temperature, relative humidity, and CO₂ are given in Table 2.

Table 2: Greenhouse climate safety bounds (temperature °C, relative humidity %, CO₂ ppm)

Variable	Lower bound	Upper bound
Temperature	15 °C	34 °C
Relative humidity	50 %	85 %
CO ₂ concentration	300 ppm	1600 ppm

109 The penalty weights are $\boldsymbol{\lambda} = [4 \times 10^{-4}, 5 \times 10^{-3}, 7 \times 10^{-4}]^T$. This
110 approach of incorporating safety bounds into policy optimization through
111 penalty terms is consistent with classical practice in constrained reinforce-
112 ment learning^{???}.

113 2.4. Task Distribution and Meta-RL Formulation

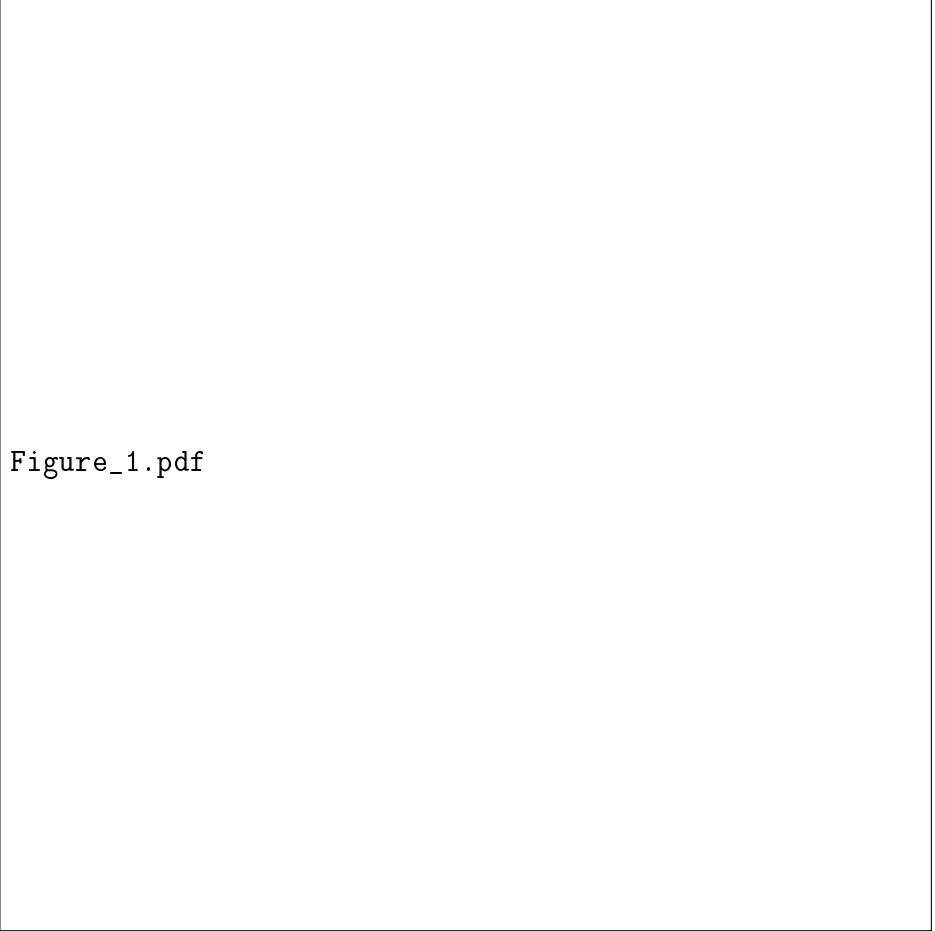
114 We view greenhouse control as a meta-reinforcement learning problem.
115 We define a single control task as $\tau = \{w, s, p, c\}$. In the formula, w is the
116 weather trajectory. s is the start date. p is the greenhouse parameters. c is

the scenario parameters. Weather year and start date cause task differences. So we use 2010 weather as the training distribution $p_{\text{train}}(\tau)$. We randomly select start dates between day 59 and day 96. We use 2011–2013 weather as the test distribution $p_{\text{test}}(\tau)$. We test on days 59, 80, and 100.

For each task segment, we define the support set $\mathcal{D}_{\tau}^{\text{sup}}$ and query set $\mathcal{D}_{\tau}^{\text{qry}}$. The support set encodes task context for the auxiliary correction module. The query set provides the trajectories on which corrected advantages are computed and used for optimization. The objective of Agri-MetaRL is Eq. (3).

$$\max_{\theta, \phi} \mathbb{E}_{\tau \sim p_{\text{train}}(\tau)} [J_{\tau}(U_{\phi}(\pi_{\theta}, \mathcal{D}_{\tau}^{\text{sup}}), \mathcal{D}_{\tau}^{\text{qry}})], \quad (3)$$

Here, U_{ϕ} denotes the support-conditioned advantage correction mechanism used during training, rather than a separate inner-loop policy optimization step. Figure 1 shows the overall system architecture, which consists of three main components and their interactions. First, a task distribution driven by weather year, start date, and scenario conditions defines a family of related control tasks. Second, the greenhouse environment receives control actions and returns observations (state) and rewards at each step. Third, the Agri-MetaRL agent comprises the support set for context encoding, recurrent Actor-Critic networks for policy and value estimation, and the meta-adaptation module for task-adaptive advantage correction. During operation, the agent observes the greenhouse state, selects actions through the policy, and receives rewards from the environment, forming a closed control loop.



Figure_1.pdf

Figure 1: Overall architecture of the greenhouse meta-reinforcement learning control system (Venlo-type greenhouse, tomato, GreenLight-Gym environment), including task distribution, greenhouse environment, Agri-MetaRL agent, and control variables such as temperature, humidity, and CO₂.

138 *2.5. Base Policy: Recurrent PPO*

139 Agri-MetaRL uses Recurrent PPO as the base learner. PPO constrains
140 policy update magnitude through a clipped objective^{? ?}. GAE provides a
141 low-variance, stable approximation for advantage estimation[?]. Recurrent

PPO further introduces LSTM to handle partial observability and temporal dependencies. This paper adopts a separated actor-critic LSTM structure. Both the policy and value networks use 1-layer, 256 hidden-unit LSTMs. The policy head fully connected layers are [1024, 1024]. The value head is [128, 128]. The learning rate is 1.161×10^{-4} , $\gamma = 0.9631$, GAE $\lambda = 0.9666$, batch size is 512, and training epochs are 8. These hyperparameters align with the GreenLight-Gym benchmark and sb3-contrib Recurrent PPO defaults, with minor adjustments from preliminary grid search to ensure comparability with baseline methods.

2.6. Agri-MetaRL: Support-Query Advantage Correction

Unlike typical Meta-RL methods, MAML learns a quickly adaptable initialization through inner-outer optimization loops. PEARL conditions the policy on latent context variables. Both require extra gradient computation or context inference modules, and policy structure can vary with task. This paper adopts a different design. On the basis of Recurrent PPO, only the advantage estimation step is corrected in a task-aware manner during training. The policy network π_θ remains single and does not switch with task. Meta-learning is embodied in how the advantage signal used for policy gradients is adjusted according to support-derived task context. This design does not require MAML-style inner update loops. It also does not inject context into policy input like PEARL. Instead, support-query partitioning conditions the auxiliary advantage correction module. In this way, the method aims to improve cross-task generalization while maintaining training-flow compatibility with standard PPO and engineering simplicity. The unique contribution of this paper is the introduction of a support-query structure into the advan-

167 tage estimation step of RL. Through the lightweight MetaAdvantageHead,
 168 the method learns how to correct advantages rather than adding task-specific
 169 policy branches.

170 Unlike traditional PPO and Recurrent PPO, which use the same GAE-
 171 style advantages for policy updates on all data, Agri-MetaRL splits each task
 172 segment into support and query sets. The support set encodes task context
 173 associated with the task variables $\tau = \{w, s, p, c\}$, including weather trajec-
 174 tory, start date, and greenhouse or scenario settings. On the query set, task-
 175 relevant correction is applied to the raw advantages. The corrected advan-
 176 tage $\tilde{A}_t = A_t + \delta_\phi(o_t, A_t, c_\tau)$ is then used for policy updates. In this way, the
 177 training signal can vary with task context. PPO and Recurrent PPO tend to
 178 learn fixed policies on single or nearly single environment distributions. Per-
 179 formance may therefore degrade when task conditions change across weather
 180 years, start dates, or related settings. Agri-MetaRL instead models green-
 181 house control as a task distribution and learns a support-conditioned ad-
 182 vantage correction rule during training. This design is intended to improve
 183 cross-task robustness without modifying the PPO backbone, adding extra
 184 policy branches, or introducing heavy inner-outer optimizers.

185 Figure 2 illustrates the workflow of the MetaAdvantageHead-based ad-
 186 vantage correction mechanism. It contains three modules.

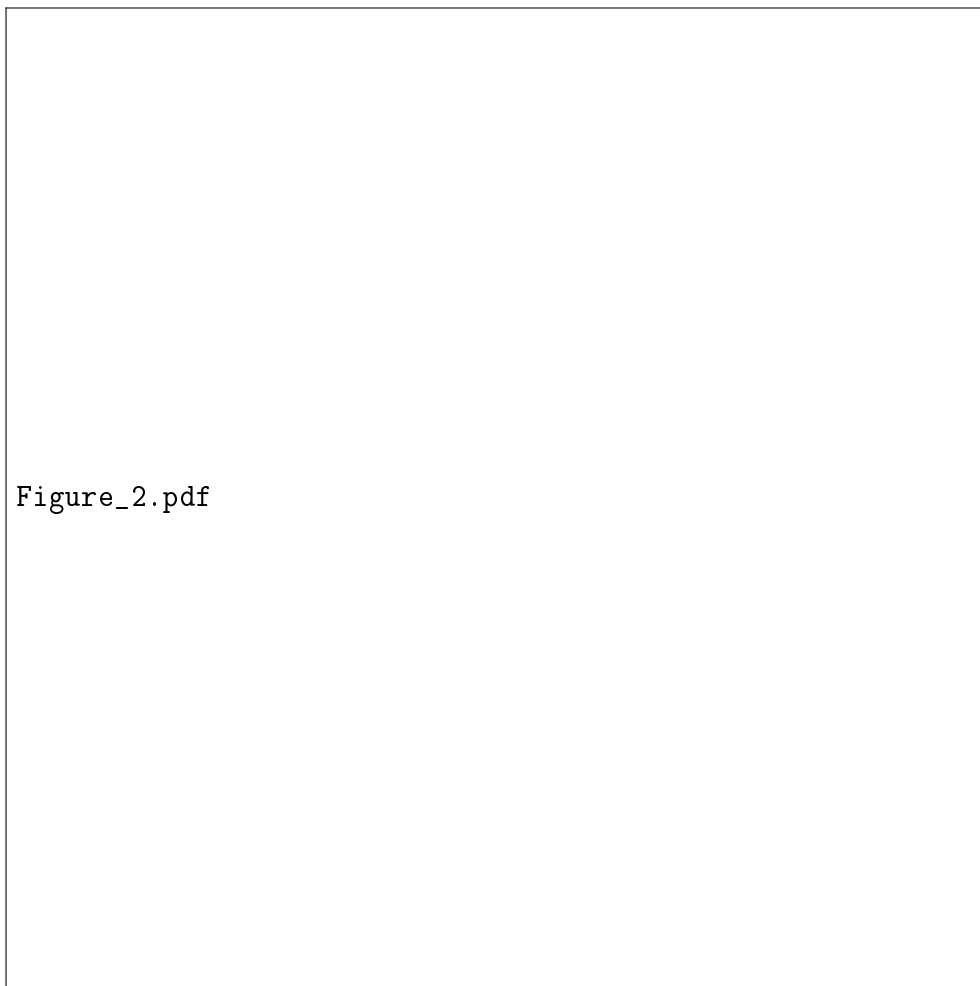


Figure 2: Workflow of MetaAdvantageHead in Agri-MetaRL (based on Recurrent PPO), including task and data partitioning, feature processing and computation, and advantage correction and update modules.

187 First, each collected task segment is partitioned into support and query
 188 parts. The support set ratio is 50% ($meta_support_ratio = 0.5$). The min-
 189 imum support steps are 256 ($meta_min_support_steps = 256$). Conceptu-
 190 ally, this corresponds to an early support portion and a later query portion

191 within the segment. Then, the support set enters the task context encoder.
 192 The encoder extracts statistics (e.g., mean and variance) from support set
 193 states, advantages, rewards, and returns, and outputs a compact context
 194 vector c_τ . Meanwhile, the query set is processed by the base Recurrent PPO
 195 module to obtain the current observation o_t and raw advantage A_t . Next, the
 196 advantage correction head receives three inputs: context vector c_τ , current
 197 observation o_t , and raw advantage A_t . It outputs a correction term through
 198 the learnable network δ_ϕ , which is added to the raw advantage:

$$\tilde{A}_t = A_t + \delta_\phi(o_t, A_t, c_\tau), \quad (4)$$

199 Its implementation contains a context encoder and an advantage correction
 200 head. The context encoder takes a 66-dimensional input, consisting of the
 201 mean and standard deviation of the first 30 observation dimensions in the
 202 support set, together with the mean and standard deviation of support-
 203 set advantages, rewards, and returns. It is implemented as two Linear-
 204 LayerNorm-SiLU blocks with dimensions $[66 \rightarrow 128 \rightarrow 64]$, and outputs a
 205 64-dimensional context vector c_τ . For each query step, the current observa-
 206 tion o_t is first projected to 64 dimensions by a linear layer. This projected
 207 observation, the raw advantage A_t (1 dim), and the context vector c_τ (64
 208 dim) are concatenated into a 129-dimensional input to the correction head.
 209 The correction head is a 2-layer MLP, $[129 \rightarrow 128 \rightarrow 1]$, with SiLU activation
 210 in the hidden layer, and outputs a scalar correction term $\delta_\phi(o_t, A_t, c_\tau)$. The
 211 corrected advantage is then normalized and clipped before being used in PPO
 212 policy updates. For meta-loss optimization, the target advantage is defined
 213 as $G_t - V_\psi(o_t)$ and standardized within the query set, so that the auxiliary

214 module learns task-sensitive corrections toward a return-based target.

$$\mathcal{L}_{meta} = \frac{1}{|\mathcal{D}_\tau^{qry}|} \sum_{t \in \mathcal{D}_\tau^{qry}} (\tilde{A}_t - \hat{A}_t)^2 + \beta \|\tilde{A}_t - A_t\|_2^2, \quad (5)$$

215 where the target advantage $\hat{A}_t$ is defined as the difference between return
216 and value estimate. It is standardized to zero mean and unit variance within
217 the query set:

$$\hat{A}_t = \frac{(G_t - V_\psi(o_t)) - \mu_{\mathcal{D}_\tau^{qry}}}{\sigma_{\mathcal{D}_\tau^{qry}} + \epsilon}, \quad (6)$$

218 $G_t = \sum_{k=0}^{T-t} \gamma^k r_{t+k}$ is the discounted return from time t , $V_\psi(o_t)$ is the Critic
219 network’s value estimate for the current observation, $\mu_{\mathcal{D}_\tau^{qry}}$ and $\sigma_{\mathcal{D}_\tau^{qry}}$ are the
220 mean and standard deviation of $G_t - V_\psi(o_t)$ within the query set, and ϵ is a
221 numerical stability constant. β is set to 0.05. This design preserves the PPO
222 base structure. By making the corrected advantage approach the return-
223 based target, the auxiliary module is encouraged to learn task-sensitive cor-
224 rections that are useful for training.

225 From a mechanistic perspective, support trajectories provide implicit in-
226 formation about the current task, including weather trajectory, start date,
227 greenhouse or scenario settings, diurnal rhythm, and crop stage. The con-
228 text encoder summarizes these signals from support-set states, advantages,
229 rewards, and returns into a compact task representation c_τ . Conditioned
230 on c_τ , the correction term $\delta_\phi(o_t, A_t, c_\tau)$ varies across tasks, so that the raw
231 advantage on the query set can be adjusted in a task-dependent manner.
232 Consequently, the corrected advantage $\tilde{A}_t$ used in PPO policy gradients re-
233 flects both the base GAE signal and support-conditioned task information.
234 Through $\mathcal{L}_{meta}$, the auxiliary module is further encouraged to move $\tilde{A}_t$ to-
235 ward the return-based target $\hat{A}_t$. If the context encoder captures features

236 that are stable across related greenhouse tasks, this mechanism can provide a
237 more task-aware training signal and may contribute to improved robustness
238 on held-out weather distributions, without introducing task-specific policy
239 branches.

240 2.7. Training and Testing Protocol

241 Total training steps are 2×10^6 . The number of parallel environments is 8.
242 Model selection uses deterministic evaluation. Approximately every 2×10^5
243 steps, evaluation is run once on the evaluation environment (training distri-
244 bution: year 2010, day 59). The checkpoint with the highest mean reward
245 is selected for subsequent evaluation. This paper uses a fixed protocol for
246 evaluation. PPO, Recurrent PPO, Agri-MetaRL, and the rule-based baseline
247 are compared on single-episode performance in a deterministic environment.
248 Multi-seed testing is conducted on both training weather distribution and
249 held-out weather distribution.

250 2.8. Experimental Design and Statistical Analysis

251 To evaluate method robustness, each algorithm was trained and evaluated
252 with 5 independent random seeds. Seeds were fixed from $\{42, 123, 456, 789, 1024\}$.
253 During training, each episode’s start date was randomly sampled within the
254 task definition range. Weather year was partitioned by training/test distri-
255 bution. During testing, deterministic environment and fixed start days were
256 used to eliminate evaluation-stage randomness. Performance comparison be-
257 tween methods is reported as mean $\pm$ standard deviation over 5 replicates.
258 Given the small sample size, this paper mainly performs descriptive statis-
259 tics. For inferential tests, paired t -test or Wilcoxon signed-rank test can be

260 used to compare return differences between methods under the same seed.

261 *2.9. Evaluation Metrics*

262 This paper mainly uses the metrics defined in this subsection for eval-
263 uation: episode total return, temperature/humidity/CO₂ violation steps,
264 cumulative economic metrics (Revenue, Heat, Electricity, CO₂, EPI), and
265 train-vs-test distribution comparison. To analyze control behavior, this pa-
266 per also presents indoor climate evolution and key actuator trajectories over
267 the full 60 d cycle.

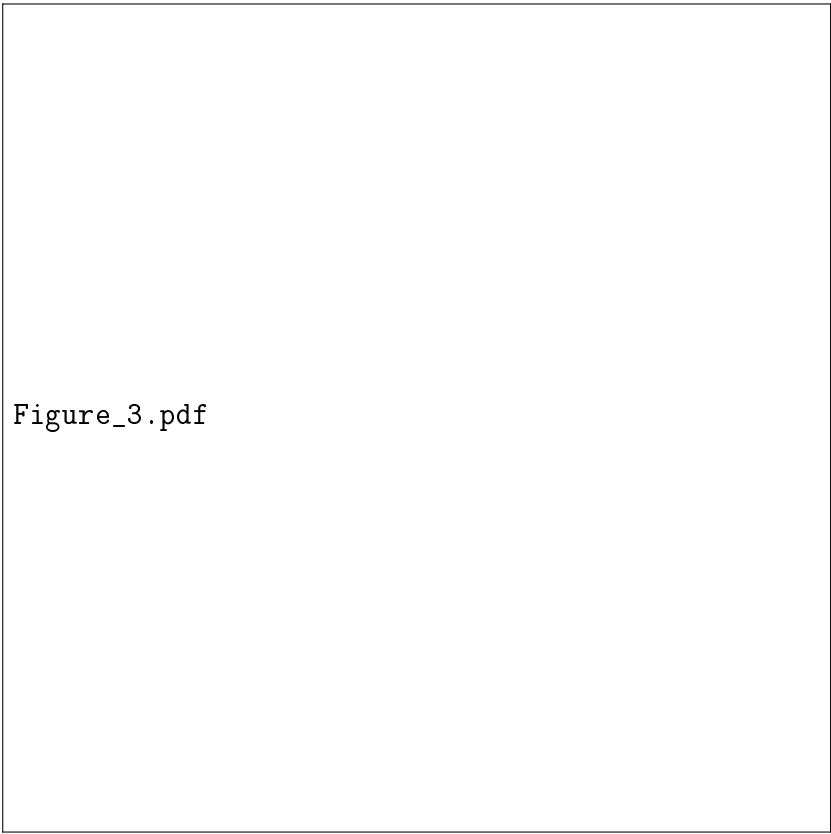
268 *2.10. Software and Reproducibility*

269 Experiments used Python 3.11, Stable-Baselines3[?] [?], sb3-contrib[?], Py-
270 Torch[?], Gymnasium[?], and GreenLight-Gym[?]. Unlike studies that only
271 use off-the-shelf software and hardware, this paper implements a custom
272 MetaAdvantageHead module on top of Recurrent PPO. This module is a
273 researcher-developed algorithm component. It is not included in the stan-
274 dard implementation of Stable-Baselines3 or sb3-contrib. To support paper
275 reproduction, this paper further provides a unified pipeline for figure and
276 table generation, enabling reproducible generation of the figures and data
277 summaries reported in the manuscript.

278 **3. Results**

279 *3.1. Learning Curves and Final Returns*

280 Figure 3 shows the mean training learning curves of PPO, Recurrent
281 PPO, and Agri-MetaRL under 5 random seeds. The shaded area indicates
282 one standard deviation across seeds.



Figure_3.pdf

Figure 3: Episode mean return versus training steps for PPO, Recurrent PPO, and Agri-MetaRL under 5 random seeds in the GreenLight-Gym greenhouse environment; shaded area indicates mean $\pm$ standard deviation across seeds.

283 Agri-MetaRL maintains a lead from the early-mid training phase. Re-
284 current PPO is consistently higher than PPO overall. Table 3 summarizes
285 the final mean returns. Agri-MetaRL reaches 3955 ± 16 . This is higher than
286 Recurrent PPO’s 3886 ± 32 and PPO’s 3811 ± 8 . These results suggest
287 that support-query advantage correction improves final performance and is
288 associated with stable behavior across random seeds.

Table 3: Comparison of final mean training returns under 5 random seeds

Method	Final mean episode return
Agri-MetaRL	3955 ± 16
Recurrent PPO	3886 ± 32
PPO	3811 ± 8

3.2. Climate Constraint Performance

Table 4 compares climate violation steps of different methods under the fixed protocol.

Table 4: Comparison of climate violation steps under fixed protocol (mean $\pm$ SD; unit: steps, 15 min/step)

Method	Temperature	Humidity	CO ₂
Agri-MetaRL	97 ± 44	98 ± 29	11 ± 5
Recurrent PPO	110 ± 55	166 ± 33	28 ± 8
PPO	118 ± 28	172 ± 68	33 ± 12

Agri-MetaRL performs best on all three dimensions: temperature, humidity, and CO₂. Violation steps are 97 ± 44 , 98 ± 29 , and 11 ± 5 , respectively. Recurrent PPO is second with 110 ± 55 , 166 ± 33 , and 28 ± 8 . PPO has 118 ± 28 , 172 ± 68 , and 33 ± 12 . Overall, Agri-MetaRL has a clear advantage in climate constraint control.

3.3. Economic Metrics Comparison

Figure 4 shows the cumulative economic metrics results.

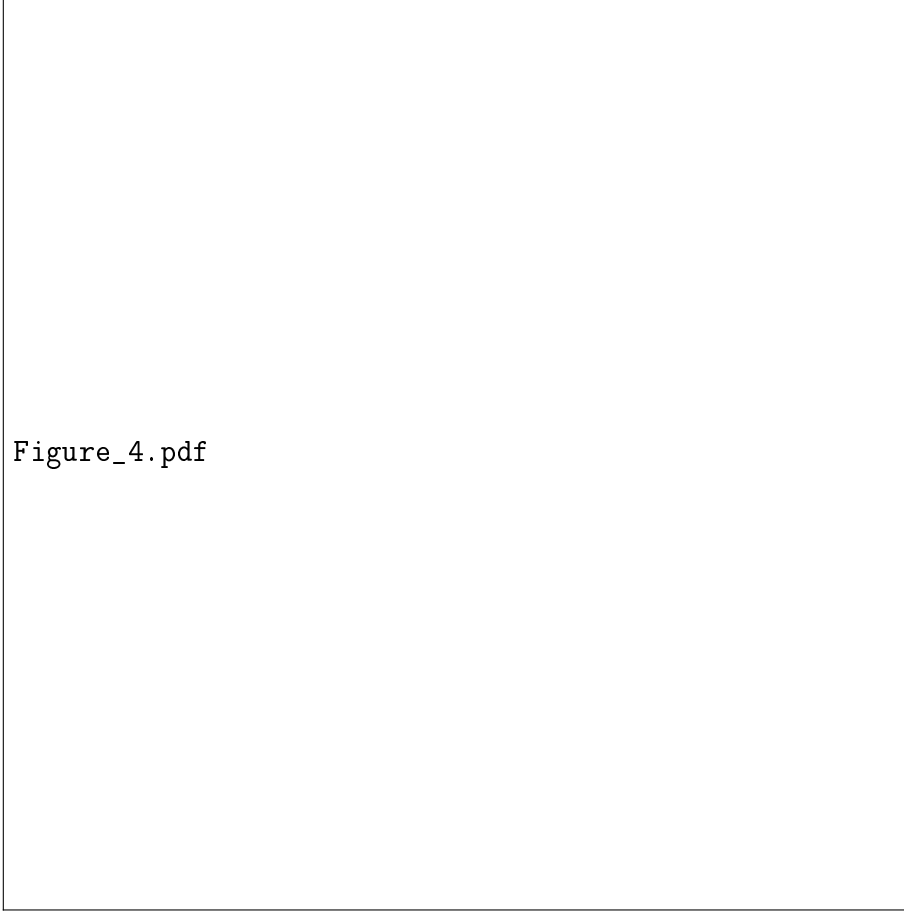


Figure 4: Comparison of cumulative economic metrics (Revenue, Heat, Electricity, CO₂, EPI; unit: €/m²) for PPO, Recurrent PPO, Agri-MetaRL, and rule-based baseline under fixed protocol (year 2010, day 59).

299 Compared with the rule-based baseline, all three RL methods significantly
300 reduce electricity and CO₂ costs and improve overall economic returns. Un-
301 der multi-seed statistics, Agri-MetaRL has Revenue of 11.56 ± 0.12 €/m²,
302 Heat of 8.12 ± 0.32 €/m², CO₂ cost of 0.08 ± 0.02 €/m², and final EPI of
303 3.46 ± 0.30 €/m². This is higher than Recurrent PPO's 3.34 ± 0.22 €/m² and

304 PPO’s 3.29 ± 0.14 €/m². These results suggest that support-query advantage
 305 correction can improve the balance between returns and resource consump-
 306 tion. The rule-based baseline has higher Revenue, but its Electricity and
 307 CO₂ costs are much larger. Therefore, its overall EPI remains significantly
 308 negative (-4.76 €/m²).

309 Table 5 summarizes the comprehensive performance of each controller on
 310 the training and held-out test distributions in the common GreenLight-Gym
 311 benchmark format. Return denotes episode total return. EPI is the Eco-
 312 nomic Production Index (€/m²). TWB_{CO₂}, TWB_T, and TWB_{RH} are the
 313 time-within-bounds percentages for CO₂, temperature, and relative humid-
 314 ity, respectively, converted from violation steps over the 5760-step episode.
 315 The training distribution used year 2010 weather with fixed start day (day
 316 59). The test distribution used 2011–2013 held-out weather with start days
 317 randomly sampled from 59/80/100.

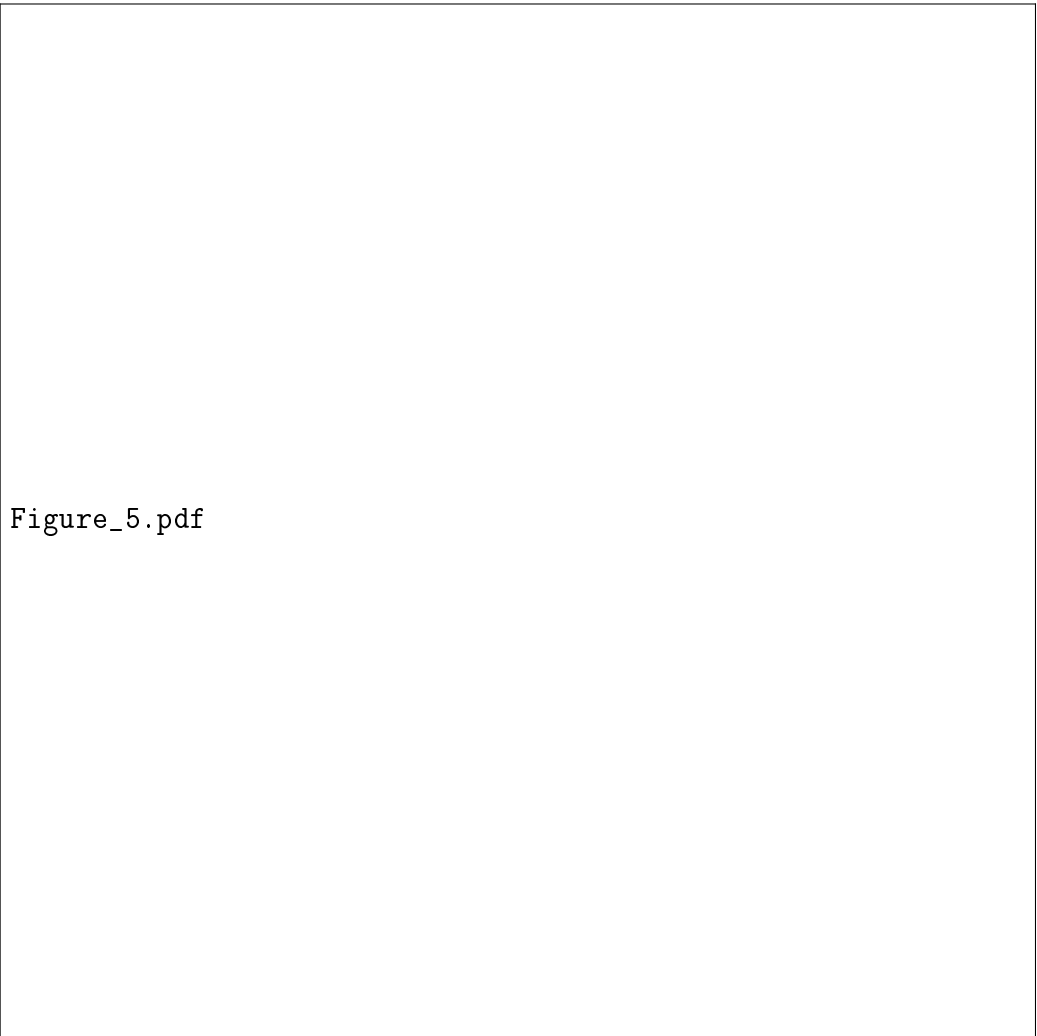
318 EPI and Return on the test distribution are both higher than on the
 319 training distribution. A likely reason is that the test set includes later start
 320 days such as day 80 and 100. These runs correspond to the mid-to-late
 321 growth season (approximately late March to early June), when day length
 322 is longer, heating demand is lower, and photosynthetic potential is higher.
 323 Amsterdam weather in 2011–2013 may also provide more favorable radiation
 324 and temperature conditions than 2010 in some periods. Therefore, the higher
 325 Test EPI should be interpreted mainly as a seasonal evaluation effect rather
 326 than a direct indicator of controller superiority.

Table 5: Comprehensive performance comparison of controllers on training and held-out test distributions

Controller	Train (D_{val})					Test (D_{test})				
	Return	EPI	TWB _{CO₂} (%)	TWB _T (%)	TWB _{RH} (%)	Return	EPI	TWB _{CO₂} (%)	TWB _T (%)	TWB _{RH} (%)
Agri-MetaRL	3955 $\pm$ 16	3.46 $\pm$ 0.30	99.8	98.3	98.3	4022 $\pm$ 83	8.34 $\pm$ 0.11	100.0	98.2	90.1
Recurrent PPO	3886 $\pm$ 32	3.34 $\pm$ 0.22	99.5	98.1	97.1	3967 $\pm$ 77	8.21 $\pm$ 0.13	100.0	98.0	88.2
PPO	3811 $\pm$ 8	3.29 $\pm$ 0.14	99.4	98.0	97.0	3921 $\pm$ 74	8.13 $\pm$ 0.20	100.0	97.5	86.3
Rule-based	3094	-4.76	100.0	92.6	36.2	—	—	—	—	—

3.4. 60 d Full-Cycle Control Behavior Analysis

To further observe the long-term regulation characteristics of the controller over the full growth window, Figure ?? shows the actual simulation trajectories of Agri-MetaRL over the full 60 d cycle under the fixed protocol (year 2010, day 59). It includes indoor temperature, relative humidity, CO₂ concentration, and temporal changes in heating, ventilation, and CO₂ injection control signals. The climate subplots show upper and lower safety bounds. This figure was produced from recorded simulation trajectories under the fixed protocol.



Figure_5.pdf

Figure 5: Indoor climate variables and actuator control trajectories of Agri-MetaRL over the 60 d growth window (actual simulation results, fixed protocol year 2010 day 59). Climate variables: temperature ($^{\circ}\text{C}$), relative humidity (%), CO_2 (ppm). Control inputs: heating, ventilation, and CO_2 injection (dimensionless 0–1). Gray dashed lines indicate upper and lower safety bounds. Horizontal axis is time step (15 min/step).

336 The trajectories show a clear diurnal pattern over the 60 d horizon. Tem-
337 perature remains within the safe range for most of the episode, with only

338 brief night-time excursions, which is consistent with the temperature vio-
 339 lation statistics reported in Table 4. CO₂ concentration is generally higher
 340 during the daytime and decreases during the night, with only a small number
 341 of violations on some seeds (11 ± 5 steps in Table 4). The control trajec-
 342 tories indicate coordinated use of heating, ventilation, and CO₂ injection.
 343 Heating increases during low-temperature or low-radiation periods, ventila-
 344 tion becomes more active under high-temperature and high-humidity condi-
 345 tions, and CO₂ injection is concentrated mainly during daytime periods when
 346 photosynthesis is active. Taken together with the learning curves, climate
 347 violation statistics, and economic metrics, these observations suggest that
 348 Agri-MetaRL maintains a balanced control behavior over the full 60 d hori-
 349 zon in terms of climate regulation, control smoothness, and return-oriented
 350 operation.

351 4. Discussion

352 The results of this paper indicate that explicitly modeling greenhouse
 353 control as a task distribution and introducing support-query advantage cor-
 354 rection on top of Recurrent PPO merits further investigation. Compared with
 355 Recurrent PPO, Agri-MetaRL achieves higher final training return under the
 356 current protocol. It also attains the highest EPI among the RL baselines and
 357 maintains competitive performance on the held-out weather distribution. Be-
 358 cause Return and EPI on the held-out distribution are higher than on the
 359 training distribution for all three RL methods, this pattern should be in-
 360 terpreted together with the seasonal differences in the evaluation protocol
 361 rather than as direct evidence of superior generalization.

Compared with existing greenhouse RL studies, Morcego et al.[?] compared DDPG with MPC on the GreenLight model and demonstrated RL potential in greenhouse climate control. Ajagekar et al.[?] combined deep RL with robust optimization for energy-efficient control of semi-closed greenhouses. Xiao et al.[?] introduced a human-machine collaborative RL framework with grower participation. These studies mostly focus on single-environment or single-policy optimization. In contrast, this paper explicitly treats greenhouse control as a task distribution over weather year, start date, and scenario conditions, and introduces support-query updates during training. Under the current protocol, the resulting policy maintains competitive performance on held-out weather distributions. GreenLight-Gym[?] provides a standardized benchmark for greenhouse RL, and this paper further constructs a unified protocol including held-out evaluation. This facilitates comparison with subsequent studies.

From the perspective of control objective decomposition, Agri-MetaRL’s main advantages are reflected in final training return and economic return (EPI). Relative to Recurrent PPO, Agri-MetaRL achieves higher return, EPI, Revenue, and better climate-constraint performance under the current protocol. In particular, final return increases from 3886 to 3955 (about 1.8%), and EPI increases from 3.34 to 3.46 €/m² (about 3.6%). Agri-MetaRL also records the lowest violation steps on temperature, humidity, and CO₂, which is consistent with the overall ranking in the learning curves and Table 5. These comparisons suggest that the support-query advantage correction mechanism can improve the balance between return and resource cost while preserving compatibility with the Recurrent PPO backbone and avoid-

387 ing heavy inner-outer optimization. Relative to PPO, the gains are larger,
388 but Recurrent PPO remains the primary baseline for interpreting the incre-
389 mental contribution of the proposed method.

390 At the same time, this paper also observes clear limitations. The results
391 are still mainly based on the current benchmark and task partitioning. Fur-
392 ther validation of robustness is needed under richer parameter perturbations,
393 longer time spans, and larger-scale multi-seed experiments. Future work can
394 continue to optimize reward design, context feature construction, and multi-
395 objective trade-offs. From a practical application perspective, Agri-MetaRL’s
396 lightweight design (no added policy branches, no heavy inner-outer optimiz-
397 ers) is conducive to integration into existing greenhouse control systems. The
398 trade-off between support set length and computational cost still needs at-
399 tention. In addition, simulation-to-reality transfer requires further validation
400 in real greenhouses, especially for sensor noise, model bias, and delay.

401 This paper should also be viewed as an engineering prototype for Meta-
402 RL-oriented greenhouse control, not as the final answer to the greenhouse
403 meta-learning problem. The current implementation already has a clear
404 support/query meta-update skeleton and a reproducible experimental and
405 visualization workflow. But to form stronger statistical evidence, extension
406 experiments are still needed with more seeds, richer task distributions, and
407 more complete multi-baseline comparisons.

408 5. Conclusion

409 This paper proposes Agri-MetaRL, an agricultural meta-reinforcement
410 learning algorithm for greenhouse climate control. The method formulates

411 greenhouse control tasks as a task distribution over weather year, start date,
412 and scenario conditions. On the basis of Recurrent PPO, it introduces the
413 support-query MetaAdvantageHead. Support-derived task context is used to
414 correct query advantages during training. Compared with traditional single-
415 environment RL, Agri-MetaRL places more emphasis on a cross-task trans-
416 ferable update structure.

417 Under the current protocol, Agri-MetaRL achieved higher final train-
418 ing returns. It also maintained relatively stable performance on the held-
419 out weather distribution. Climate constraint analysis and economic metrics
420 comparison further support the effectiveness of task-context encoding and
421 advantage correction. Overall, the results indicate that Agri-MetaRL pro-
422 vides a practical and extensible framework for Meta-RL-oriented greenhouse
423 climate control research.

424 **Data Availability**

425 The source code and evaluation pipeline for Agri-MetaRL are available
426 at <https://github.com/1240945123/Agri-MetaRL>.

427 **Author contributions: CRediT**

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442 The authors declare that they have no known competing financial inter-
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444 reported in this paper.